

L10 — Animation I

Principles, anatomy, kinematics, and skinning

Study summary generated from lecture slides. ETH AI for Digital Characters.

Character fundamentals

- Stylized vs realistic: stylized avoids uncanny valley
- Uncanny valley: appeal drops as human-likeness increases (before real human)

Disney's 12 principles (know for exam)

- 1. Squash and Stretch - mass and rigidity
- 2. Anticipation - preparation before main action
- 3. Staging - clear focus
- 4. Straight Ahead vs Pose to Pose
- 5. Follow Through and Overlapping Action
- 6. Slow In and Slow Out - ease in/out (NOT linear)
- 7. Arc - natural curved paths
- 8. Secondary Action
- 9. Timing
- 10. Exaggeration
- 11. Solid Drawing
- 12. Appeal

Anatomy and rigging

- Facial muscles: bone-to-skin; wrinkles perpendicular to contraction
- Joints: hinge, ball-and-socket, etc.; 0-6 degrees of freedom
- Skeleton hierarchy: bones define movement, joints define rotation

Kinematics

- Forward kinematics (FK): joint angles $\rightarrow$ end-effector position
- Inverse kinematics (IK): target position $\rightarrow$ joint angles
- Jacobian method: $\Delta\theta = J^+ \cdot \Delta e$
- Pole vector: controls elbow/knee bend direction in IK
- Mass-spring systems for secondary motion

Skinning

- Linear Blend Skinning (LBS): $v' = \sum w_{ij} \cdot (R_j \cdot v_i + T_j)$
- Each vertex influenced by multiple bones with weights summing to 1
- Candy-wrapper artifact: unnatural twisting at large joint angles
- Motion capture: blend weights per vertex